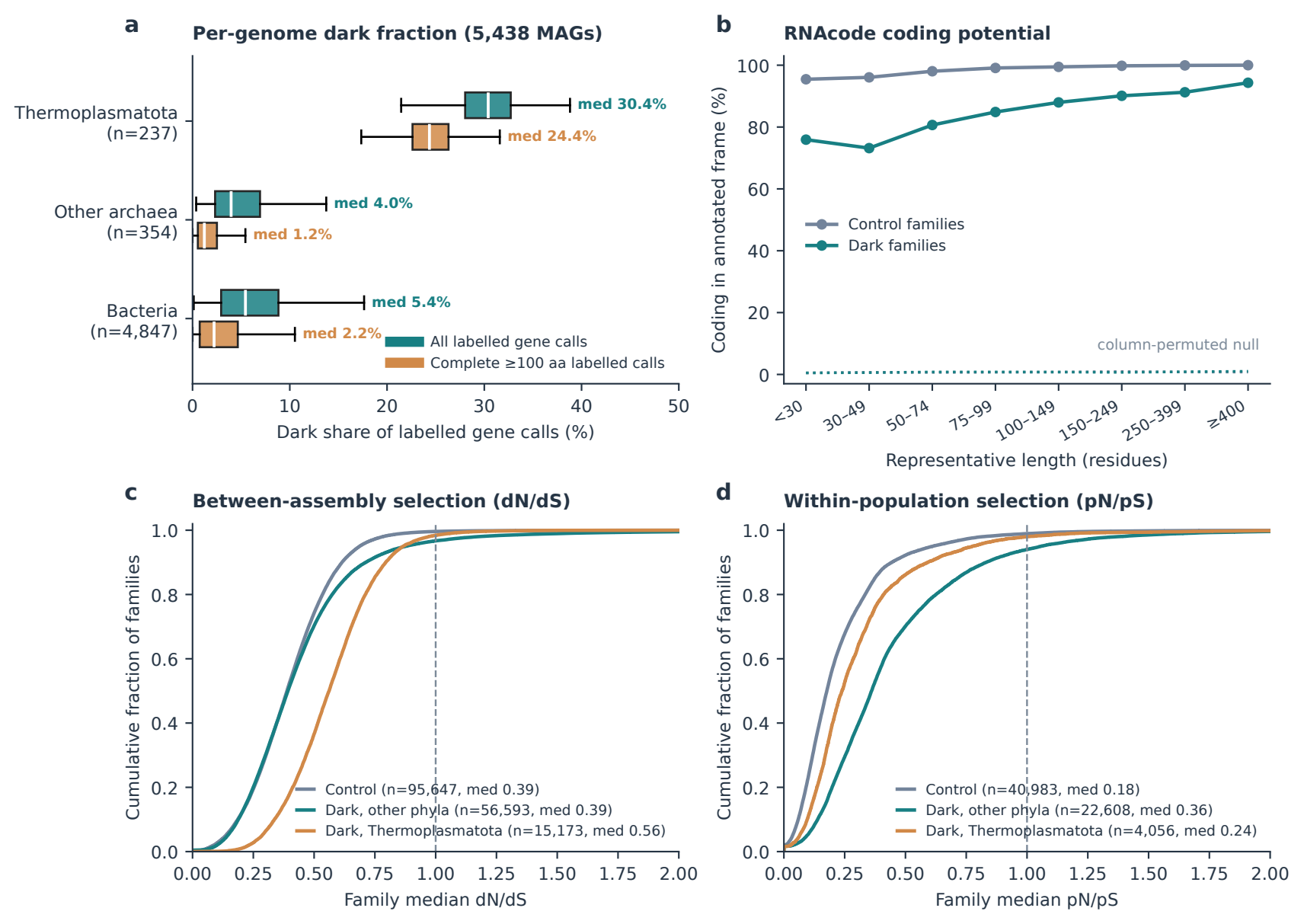

a
Per-genome dark fraction (5,438 MAGs)
Thermoplasmatota (n=237)
med 30.4%
med 24.4%
Other archaea (n=354)
med 4.0%
med 1.2%
Bacteria (n=4,847)
med 5.4%
med 2.2%
All labelled gene calls
Complete ≥100 aa labelled calls
0
10
20
30
40
50
Dark share of labelled gene calls (%)
b
RNAcode coding potential
Coding in annotated frame (%)
0
20
40
60
80
100
Control families
Dark families
column-permuted null
<30
30–49
50–74
75–99
100–149
150–249
250–399
≥400
Representative length (residues)
c
Between-assembly selection (dN/dS)
Cumulative fraction of families
0.0
0.2
0.4
0.6
0.8
1.0
Control (n=95,647, med 0.39)
Dark, other phyla (n=56,593, med 0.39)
Dark, Thermoplasmatota (n=15,173, med 0.56)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Family median dN/dS
d
Within-population selection (pN/pS)
Cumulative fraction of families
Control (n=40,983, med 0.18)
Dark, other phyla (n=22,608, med 0.36)
Dark, Thermoplasmatota (n=4,056, med 0.24)
Family median pN/pS